ACCESS Discover Request for Computational Resources

Title of Project: Evaluation of ADMETS Properties of Selected Chemical Products for Drug Discovery

Name and institution: Quest Student Research Institute (Quest SRI)

Project Lead: Dr. Henry David Snyder, retired, Gallaudet University; Dr. Tugba Kucukkal, PI, Gallaudet University

Research and Education Objectives: The objective of this project is to explore ADMETS (absorption, distribution, metabolism, excretion, toxicity and stability) properties of identified natural chemical products that are candidate Plk1 inhibitors and their structural derivatives. The following five natural products will be studied: luteolin, resveratrol, xanthatin, chaetoglobosin e, quercetin. This research group will be composed of approximately 10 high school students. Each will be assigned 10-20 molecules to investigate and they will be responsible for following group protocols for recording and processing their data in multiple, shared Google spreadsheets. Participants will be trained about how to read scientific papers, how to conduct a scientific literature search, what are the basics and importance of computational chemistry and how to properly run and interpret output from various software applications. Participants will become familiar with the supercomputer environment and employ high-speed computers to perform optimization and vibrational frequency calculations for their assigned molecules. This year, our research team accepted two students through the ACS Project SEED Program (Summer Experiences for the Economically Disadvantaged). This program supports economically disadvantaged high school students who completed a chemistry course.

Research Plan: Students will first form a database of small molecules composed of the five identified products and selected structurally-related derivatives. This will be done by searching each of the five natural products in the ZINC20 [1] (ZINC Is Not Commercial 2020) database of commercially available compounds and downloading the structure files of selected molecules that are identified as closely related by ZINC20. COCONUT [2] may be used as an additional reference. After that they will explore the ADMETS properties of molecules in this newly formed database. Software including ADMETlab 2.0, SwissADME and Toxtree will be utilized to assess their medical and drug relevant properties. Avogadro will be used for visualizations. Gaussian16 will be used for quantum mechanical calculations. The results will be collected in shared Google Sheets files for post-processing. Each of the ADMET software applications computes its own drug characterization. The overlap between ADMETlab, SwissADME and Toxtree may provide multiple viewpoints. All candidate molecules will be ranked according to the ADMETS results to select molecules for investigation in the future with more in depth molecular trials such as molecular dynamics. Since this project is an early stage of a larger, longer duration drug discovery effort, it is important to organize results in a form that can be interpreted and reprocessed in conformance with priorities of follow-on research groups. A similar research plan was used in 2022 [3].

Description of Resource Needs: High-speed computation resources are needed to evaluate properties of molecules in a timely manner. All participants will work primarily on their personal computers. To make this experience as equitable as possible Quest SRI provides economically disadvantaged students with high speed internet and a personal computer for the duration of the internship. Success of this early molecule screening project is dependent on identifying a wide sample (over 100) of likely candidate molecules. Student researchers will join regular group meetings with Dr. Snyder on Zoom and will remain in contact through email at other times. This Quest internship lasts 6 weeks and quantum mechanical calculations with Gaussian [4] software of each molecule using the DFT-6-31G* level of density function theory [5] will take days to weeks on personal computers, depending on the size of the

molecule. Given the limited time available to researchers, high-speed computer resources are needed. Considerable supercomputer time and modest storage (mostly kb-size input and output text files) is needed for Gaussian16 processing.. Therefore, the supercomputing time provided through NSF ACCESS is of outmost importance for the success of this research project. The primary goal of this computation is to determine the HOMO-LUMO energy gap between molecular orbitals. This will be used along with the ADMET features to characterize candidate molecules *in silico*.

[1] J. Chem. Inf. Model. 2020, 60, 12, 6065–6073; ZINC20—A Free Ultralarge-Scale Chemical Database for Ligand Discovery; Publication Date: October 29, 2020; https://doi.org/10.1021/acs.jcim.0c00675

[2] Sorokina, M., Merseburger, P., Rajan, K. *et al.* COCONUT online: Collection of Open Natural Products database. *J Cheminform* 13, 2 (2021). https://doi.org/10.1186/s13321-020-00478-9

[3] https://quest-internship.s3.amazonaws.com/internship-links-2022.html. Note that in 2022, work with Gaussian16 was started in week 4. It was originally planned to start in week 2 to allow for long molecule processing times. Due to a delay, researcher TACC accounts were no available until week 4.

[4} https://gaussian.com/g16glance/

[5] https://gaussian.com/dft/